FIFTH SEMESTER (CUFYUGP) DEGREE EXAMINATION: MODEL PAPER 1

MATHEMATICS

ABSTRACT ALGEBRA I

Module I: Binary Operations and Isomorphic Binary Structures

Maximum Time: 2 Hours		**Maximum Marks: 70**
Section A		
All Questions can be answered. Each Question carries 3 marks (Ceiling: 24 Marks)		
1	Let $*$ be defined on $\mathbb{Q}$ by $a * b = a/b$. Does $*$ define a binary operation on $\mathbb{Q}$? Justify your answer.	**3**
2	Show that the set of all odd integers is not closed under ordinary addition.	**3**
3	Let $*$ be defined on $\mathbb{Z}$ by $a * b = \max\{a, b\}$. Show that $*$ is commutative and associative.	**3**
4	Let $*$ be defined on $\mathbb{R}$ by $a * b = \frac{a+b}{2}$. Is $*$ commutative? Is it associative?	**3**
5	Let S be a set with more than one element and let $a * b = a$ for all $a, b \in S$. Show that $*$ is associative but not commutative.	**3**
6	Give an example of a set with a binary operation which is commutative but not associative.	**3**
7	Show that the identity element of a set with a binary operation, if it exists, is unique.	**3**
8	Let $*$ be defined on $\mathbb{Z}$ by $a * b = a + b + 1$. Find the identity element of $\langle \mathbb{Z}, * \rangle$, if it exists.	**3**
9	State the condition that a function $f : \langle S, * \rangle \to \langle S', \circ \rangle$ must satisfy to be an isomorphism of binary structures.	**3**
10	Show that the binary structures $\langle \{0, 1\}, * \rangle$ with $a * b = a$, and $\langle \{0, 1\}, \circ \rangle$ with $a \circ b = \max\{a, b\}$, are not isomorphic.	**3**
Section B		
All Questions can be answered. Each Question carries 6 marks (Ceiling: 36 Marks)		
11	Let $*$ be defined on $\mathbb{Z}^+$ by $a * b = \min\{a, b\}$. Is $*$ a binary operation on $\mathbb{Z}^+$? Is it commutative? Justify your answer.	**6**
12	Show that $f : \mathbb{Z} \to 3\mathbb{Z}$ given by $f(m) = 3m$ is an isomorphism from $\langle \mathbb{Z}, + \rangle$ onto $\langle 3\mathbb{Z}, + \rangle$.	**6**
13	Show that the binary structure $\langle \mathbb{R}, + \rangle$ with operation the usual addition is isomorphic to the structure $\langle \mathbb{R}^+, \cdot \rangle$ where $\cdot$ is the usual multiplication.	**6**
14	Consider the following two binary structures: (a) $\mathbb{Z}$, the set of integers with ordinary addition; and (b) $2\mathbb{Z} = \{2n \mid n \in \mathbb{Z}\}$, the set of even integers with ordinary addition. Show that the above two binary structures are isomorphic.	**6**
15	Let $f : \langle S, * \rangle \to \langle S', \circ \rangle$ be an isomorphism of binary structures. Show that $f^{-1} : S' \to S$ is also an isomorphism of binary structures.	**6**
16	Let $f : \langle S, * \rangle \to \langle S', \circ \rangle$ be an isomorphism of binary structures. Show that if $*$ is commutative, then $\circ$ is commutative.	**6**
17	Show that the relation '$\langle S, * \rangle$ is isomorphic to $\langle S', \circ \rangle$' is an equivalence relation on binary structures.	**6**
18	Let $*$ be defined on $\mathbb{Z}$ by $a * b = a + b + 1$. Show that $\langle \mathbb{Z}, * \rangle$ is isomorphic to $\langle \mathbb{Z}, + \rangle$.	**6**
Section C		
Answer any ONE. Each Question carries 10 marks (1x10=10 Marks)		
19	Let $*$ be defined on $\mathbb{R}$ by $a * b = a + b - ab$. (a) Show that $*$ is a binary operation on $\mathbb{R}$. (b) Show that $*$ is commutative and associative. (c) Find the identity element of $\langle \mathbb{R}, * \rangle$ and show that it is unique.	**10**
20	Show that the binary structures $\langle \mathbb{R}, + \rangle$ and $\langle \mathbb{R}, \cdot \rangle$ are not isomorphic.	**10**